

	A	B	C	D	E	F	G	H	I	J	K	L
1	Standard deviation and coefficient of variation											
2	Pizza price example											
3												
4		NY Dollars					Dollars		Pesos			
5		\$	1.00	MXN	18.81		Mean	\$ 5.50	MXN	103.46		
6		\$	2.00	MXN	37.62		Sample variance	\$ ² 10.72	MXN ²	3793.69		
7		\$	3.00	MXN	56.43		Sample standard deviation	\$ 3.27	MXN	61.59		
8		\$	3.00	MXN	56.43		Sample coefficient of variation	0.60		0.60		
9		\$	5.00	MXN	94.05							
10		\$	6.00	MXN	112.86							
11		\$	7.00	MXN	131.67							
12		\$	8.00	MXN	150.48							
13		\$	9.00	MXN	169.29							
14		\$	11.00	MXN	206.91							
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The coefficient of variation tells us how much data spreads out relative to it's average. It's given by the equation below.

It's purpose is to allow comparisons of different samples or populations e.g. the spread of weight around the mean of mouse and elephants.

$$CV = \left(\frac{\sigma}{\mu} \right) \times 100\%$$